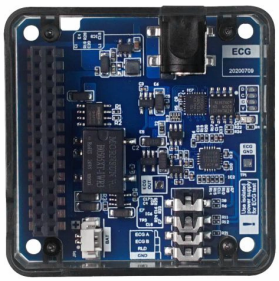
Module13.2 ECG

SKU:M034



















Description

Module13.2 ECG is a heart rate measurement module capable of being used for human heart rate statistics and ECG waveform generation. Electrocardiography (ECG) is a technique that uses an electrocardiograph to record the electrical activity of the heart during each cardiac cycle from the body surface. On the signal acquisition side, the Module13.2 ECG integrates the AD8232 single-lead heart rate monitoring front end. The collected ECG signal is processed through the AD8603 low-pass filter, then fed into a 10-bit ADC (AD7476) for analog-to-

digital conversion and input into an STM32 (which has a built-in heart rate statistics algorithm) for signal analysis. Finally, the processing results are output in serial communication format, making it easy for the host device to acquire and display the data. In terms of signal output, it adopts a full isolation design between the front-end and the digital part to enhance stability and safety.

Note

Power Supply Note

This product only supports **5V** power input. Please strictly follow this power input standard when using the product to avoid damage to the device or potential injury.

Features

- ADI integrated front-end (High signal gain G=100, with DC blocking capability)
- Serial data output
- Built-in heart rate statistics algorithm
- Full isolation between front-end and digital section
- Ultra-precision operational amplifier

Includes

- 1 x Module13.2 ECG
- 3 x ECG Lead Wires
- 6 x Lead Electrodes

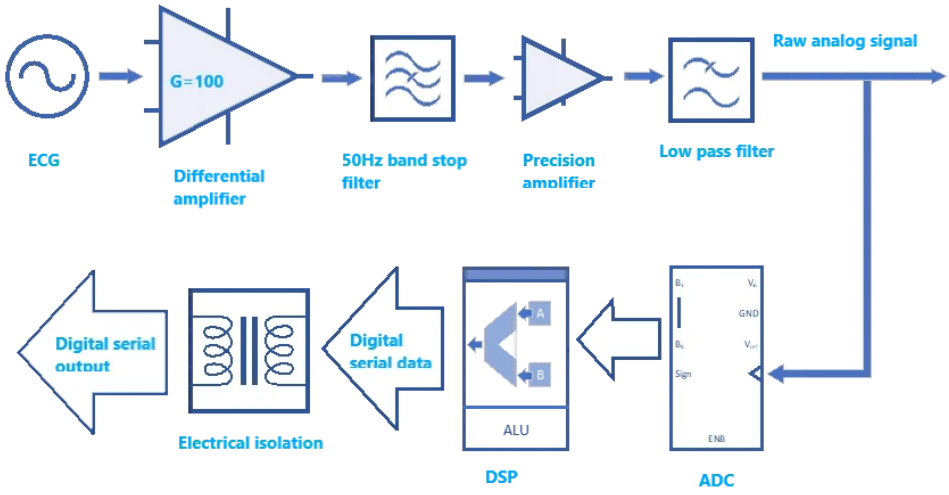
Applications

- Bioelectrical signal acquisition
- Portable ECG
- Fitness and exercise heart rate monitors

Specifications

Specification	Parameter
MCU	STM32F031G4U6
Communication	UART 115200bps
AD8232	High signal gain G=100
AD7476	10bit-ADC
Product Size	54.0 x 54.0 x 13.2mm
Product Weight	18.0g
Package Size	105.0 x 65.0 x 40.0mm
Gross Weight	101.0g

Schematics



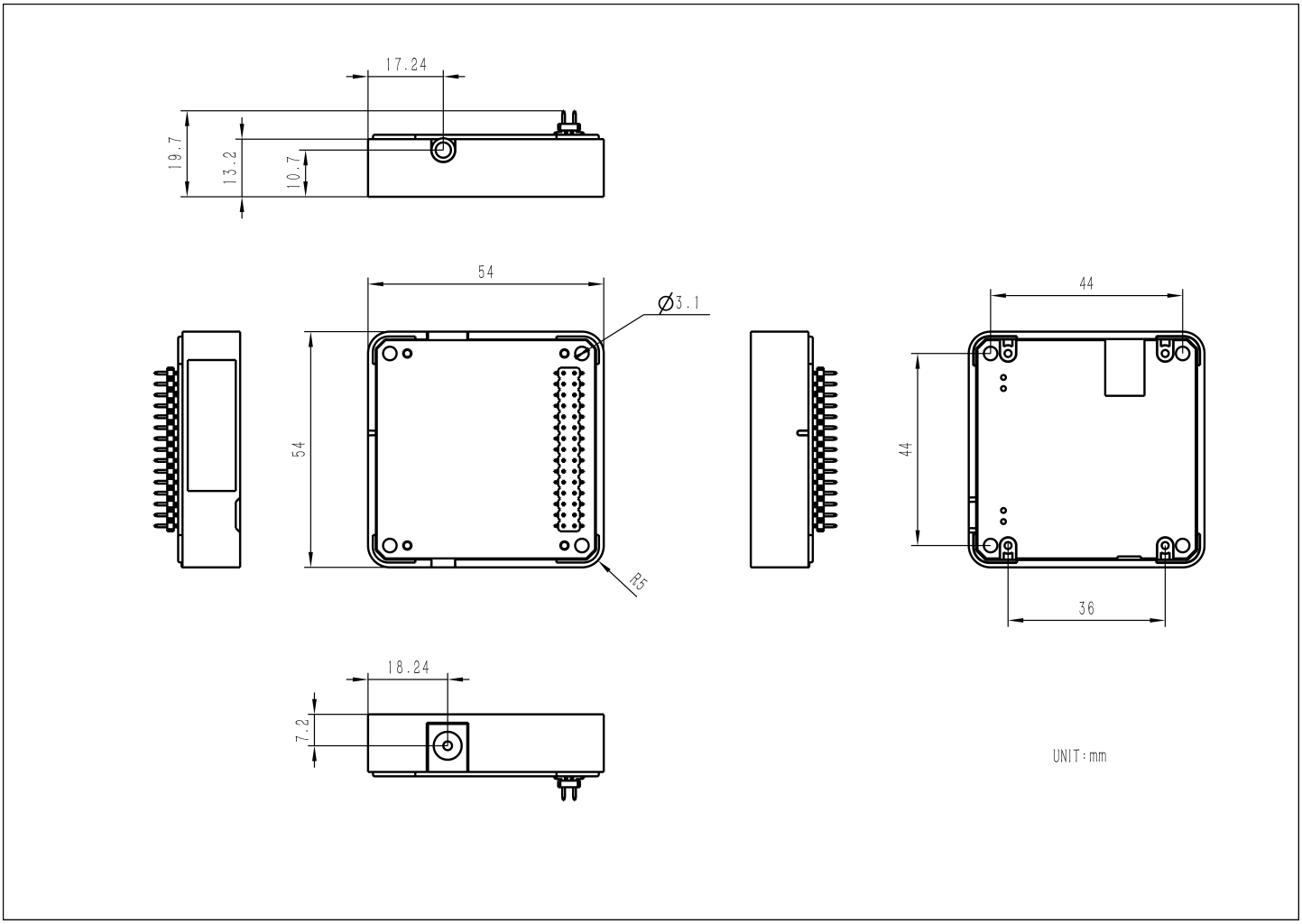
PinMap

M5-Bus

PIN	LEFT	RIGHT	PIN
GND	1	2	
GND	3	4	
GND	5	6	
	7	8	
	9	10	
	11	12	3V3
	13	14	
	15	16	
	17	18	
	19	20	RXD
	21	22	TXD
	23	24	
HPWR	25	26	
HPWR	27	28	5V
HPWR	29	30	BAT

Model Size

- Module13.2 ECG Model Size PDF



Datasheets

- AD8232
- AD8603
- AD7476
- ECG Data Packet Format

Softwares

Arduino

- Module13.2 ECG Example with M5Core

UiFlow2

- Module13.2 ECG UiFlow2 Docs

Easyloader

Easyloader	Download	Note
Module13.2 ECG Example Easyloader with M5Core	download	/

Other

PC Software Usage Instructions

When using the PC heart rate reading tool, the device needs to be flashed with the serial port transparent transmission firmware (before uploading, please change the serial port initialization pins to the actual connection pins), and the data will be forwarded to the PC.

- RawDisplay-PC Application
- M5Core UART Transparent Transmission Program

Video

- Heart rate statistics, ECG waveform display

ECG.mp4